MOHAMED KHALID M JAFFAR

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SUMMARY

Robotics and autonomy researcher with 10 years of experience across diverse robotic platforms, including ground, aerial, and legged robots, UAVs, and rockets. I specialize in motion planning, control, optimization, and AI/ML for dynamical systems, with a focus on **safety-critical decision making** for real-world autonomous agents. I am seeking R&D roles where I can contribute to high-impact, intellectually challenging projects.

EDUCATION

University of Maryland, College Park, USA — **Expected Spring '26**
Doctor of Philosophy (PhD) in Aerospace Engineering — **GPA: 3.62/4**

Indian Institute of Technology Madras, India — **2013 – 2018**
Dual Degree (B.Tech & M.Tech) in Aerospace Engineering — *Class Rank* **1**
Minor in Robotics — **GPA: 9.06/10**

SKILLS AND EXPERTISE

- **Conceptual Expertise**: Robot Autonomy, Motion Planning and Control, Optimization, Applied Mathematics, Machine Learning (ML), and Vision-Language Models (VLMs)
- **Math Background**: Linear Algebra, Functional Analysis, Probability and Stochastic Processes, Differential Geometry, Data Structures and Algorithms
- **Programming Languages**: C++, Python, Object-Oriented MATLAB, C
- **Tools and Software**: ROS, Simulink, YALMIP, CasADi, CVXPY, gurobipy, PyTorch, SolidWorks

AWARDS AND HONORS

- Ann G. Wylie **Dissertation Fellowship** for pursuing *semester-long* academic research (Fall 2024)
- **Dean's fellowship** from the University of Maryland, College Park (2019)
- **Gold Medalist** for **best overall Academic record** in Dual Degree Aerospace Engineering (2018)
- **NTU-India connect Scholarship** from NTU, Singapore; *1 among 20 students from India* (2016)

RELEVANT EXPERIENCE

Research Scientist Intern | Control for Autonomy, Mitsubishi Electric Research Labs — **2025–present**
Project: *Adaptive spatio-temporal monitoring using constrained mobile robots* | Mentor: Dr. Abraham Vinod

- Developed a heterogeneous multi-robot environmental monitoring system using aerial and ground robots that coordinate under kinematic, collision-avoidance, and battery constraints
- Implemented data-driven adaptation via Gaussian process-based uncertainty quantification and centralized multi-agent route planning using integer programming

Graduate Research Assistant | Motion and Teaming Lab, University of Maryland — **2018–25**
PhD Thesis: *Safe feedback motion re-planning of mobile robots in dynamic spaces* | Advisor: Dr. Michael Otte

- Developed provably-safe motion control algorithms for mobile robots that respect their kinodynamic constraints and compute optimal motion plans online in dynamic environments
- Designed fault-tolerant control strategies for multirotor aerial vehicles with one or more rotor failures
- Computed safe reachable sets using dynamical systems theory and numerical, non-convex optimization

Summer Scholar | U.S. Army Research Lab & GAMMA Lab, University of Maryland — **Summer '24**
Project: *Long-range navigation of legged robots in outdoor terrain* | Advisor: Dr. Dinesh Manocha & Jason Pusey

- Field experiments of quadruped autonomous navigation in unmapped forest with trees, tall grass, and mud

- Implemented robot gait adaptation to tread uneven terrains by leveraging visual perception through LLMs/VLMs, and haptic perception using inertial sensors

Masters' Research | RAFT Lab and Robotics Lab, IIT Madras **2017–18**
Thesis: *Maneuver control of quadrotors in indoor spaces* | Advisor: Dr. Ranjith Mohan & Dr. Asokan Thondiyath

- Developed on-board model-based control for aggressive maneuvers of quadrotors in tight indoor spaces
- Synthesized and compared performance of different nonlinear controllers: backstepping, PI–PID, SMC
- Carried out hardware tests for system identification and evaluation of the various control strategies

Autonomy Lead & Co-founder | Team Noctua, DeTect Technologies | Tech startup **2015–17**
Product: *Autonomous aerial robot operations for infrastructure inspection in oil-gas industrial plants*

- One of the founding members of then seed-funded startup, focusing on technology development and operations; currently valued at $138 million in Series B
- Developed autonomy algorithms for aerial robots to trace predefined routes with precision, for inspecting faults and leaks in pipelines, furnaces, and chimneys of petrochemical plants

Undergraduate Research Intern | IIT Madras & Defense R&D Organisation (DRDO) **Fall 2015**
Project: *Autonomous target-tracking unmanned aerial vehicle (UAV)* | Advisor: Dr. Joel George

- Mathematically modeled a multirotor UAV system in MATLAB; developed guidance and control strategies to generate and track reference trajectories; conducted hardware experiments using a VTOL hexacopter

APPLIED PROJECT EXPERIENCE

Systems Lead | Airbrake subteam, Terrapin Rocket team | Spaceport America Cup **2021–24**
Project: Development of drag-modulating airbrake system to control a rocket's ascent | Las Cruces, NM
Awards: Placed **category 2nd** in 2023 and **overall 1st** in 2024, out of **150 teams globally**

- Led a 6 student team across two years, developing electro-mechanical modules of deployable braking flaps
- Implemented an MPC and EKF-based autonomy stack in C++ for real-time altitude regulation of a rocket

Independent Research | Creative Engineering Project, IIT Madras **Fall 2017**
Project: Autopilot flight controller board for multirotor UAVs | Advisor: Dr. Ranjith Mohan

- Designed and built a custom embedded autopilot board integrating microcontrollers and MEMS sensors
- Achieved real-time control and stabilization, with low-level interfacing in RTOS and assembly language

Aerostructures Lead | Nimbus, IIT Madras competition team | SAE Aero Design **2016–17**
Project: Design and fabrication of radio-controlled (RC) fixed-wing airplane | Fort Worth, TX
Recognition: Placed **12th worldwide** and **4th in India**, out of **75 teams globally**

- Carried out analysis of aerodynamic characteristics, stability, and performance of the fixed-wing UAV
- Designed the aircraft structure using SolidWorks and manufactured using composites

Summer Intern | Flight Mechanics & Control Lab, NTU Singapore **Summer 2016**
Project: Hardware design and manufacturing of tiltrotor bicopter experiment setup | Advisor: Dr. Erdal Kayacan

- Designed a bicopter planform with dual-axis tilt and a 3-axis articulated test rig using SolidWorks
- Fabricated the various prototypes using composites, additive manufacturing, and metal machining

SELECTED PUBLICATIONS

Journal Articles

- **M.K.M Jaffar** and M. Otte. "Numerical computation of safe reachability sets for nonlinear dynamical systems using Lyapunov theory and SOS programming." **(in preparation)**
- **M.K.M Jaffar** and M. Otte. "Online feedback motion re-planning in dynamic spaces using invariant funnels." *International Journal of Robotics Research* **(preprint)**

- E. Carrillo, S. Yeotikar, S. Nayak, **M.K.M Jaffar**, S. Azarm, J. Herrmann, M. Otte, and M. Xu. "Communication-aware multi-agent metareasoning for decentralized task allocation." *IEEE Access* **(2021)**
- S. Nayak, S. Yeotikar, E. Carrillo, E. Rudnick-Cohen, **M.K.M Jaffar**, R. Patel, S. Azarm, J. Herrmann, M. Xu, M. Otte. "Experimental comparison of decentralized task allocation algorithms under imperfect communication." *IEEE Robotics and Automation Letters* **(2020)**

Peer-Reviewed Conference Proceedings

- **M.K.M Jaffar**, S. Di Cairano, A. Vinod. "Adaptive spatio-temporal monitoring using constrained mobile robots." *Conference on Decision and Control* **(in preparation)**
- G. Seneviratne, K. Weerakoon, M Elnoor, V. Rajgopal, H. Varatharajan, **M.K.M Jaffar**, J. Pusey, D. Manocha. "Cross attention-based multimodal representation fusion for parametric gait adaptation in complex terrains." *International Conference on Intelligent Robots and Systems* **(2025)**
- M. Elnoor, K. Weerakoon, G. Seneviratne, R. Xian, T. Guan, **M.K.M Jaffar**, V. Rajagopal, D. Manocha. "Robot navigation using physically grounded vision-language models in outdoor environments." *International Conference on Robotics and Automation* **(2025)**
- K. Weerakoon, M. Elnoor, G. Seneviratne, V. Rajagopal, S. Arul, J. Liang, **M.K.M Jaffar**, D. Manocha. "Behavioral rule guided autonomy using VLMs for robot navigation in outdoor scenes." *International Conference on Robotics and Automation* **(2025)**
- H. Yang, **M.K.M Jaffar**, and M. Otte. "Trajectory tracking while stabilizing an inverted pendulum on a quadcopter using adaptive model-predictive control." *AIAA SciTech* **(2024)**
- E. Bregin and **M.K.M Jaffar**. "Development of a drag-modulating feedback system to control a rocket's ascent." *AIAA SciTech* **(2024)**
- J. Cho, C. Kim, **M.K.M Jaffar**, M. Otte, and S. Kim. "Low-level controller in response to changes in quadrotor dynamics." *International Conference on Robotics and Automation* **(2023)**
- **M.K.M Jaffar** and M. Otte. "PiP-X: Funnel-based online feedback motion planning/replanning in dynamic environments." *Algorithmic Foundations of Robotics* **(2022)** [*also appears as a book chapter*] [***best paper award consideration***]
- **M.K.M Jaffar**, M. Velmurugan, and R. Mohan. "A novel guidance algorithm and comparison of nonlinear control strategies applied to an indoor quadrotor." *Fifth Indian Control Conference* **(2019)**

Open-source software releases

- **M.K.M Jaffar**. "`computeSOSFunnels`: A MATLAB framework for invariant funnel synthesis using Sum-of-Squares optimization." *v1.0.0 - GitHub* **(2025)**

A full list of my research **publications** is available online at **Google Scholar**.

TEACHING AND MENTORING EXPERIENCE

- **Head Instructor, CAD Lab**: Taught 150 undergraduate students core computer-aided design concepts through hands-on SolidWorks training; managed & supervised 4 teaching fellows **(UMD, Spring 2025)**
- **Mentored 5+ undergraduates** in Motion & Teaming Lab, Department of Aerospace, UMD
  Students mentored: Matthew Nam, Han Yang, Rishi Parikh, Robert Baratta, Matthias Kim **(2022–24)**
- **Teaching Assistant (TA)** for courses: Flight simulator and controls, and Fluid mechanics. Conducted lab sessions and evaluated reports, quiz papers of 60 sophomore students **(IIT Madras, 2017–18)**

PROFESSIONAL SERVICE AND LEADERSHIP

- **Academic and Professional Chair** of Graduate Student Advisory Committee (GSAC), Aerospace Department, UMD College Park **2021–22**
- **Student representative** on the **Executive Committee** of Maryland Robotics Center **2020–21**
- **Strategist and Mentor** for student teams in **Robotics**, **Electronics**, and **Aero Clubs** at the Centre for Innovation (CFI), IIT Madras **2016–18**